

Experiment 3: Implementation of Decision Tree and Random Forest for Classification on Heart Disease Dataset

1. Dataset Source

The dataset used in this experiment is the **Heart Disease Dataset**.

Source:

<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

The dataset contains medical attributes used to predict the presence of heart disease in a patient.

2. Dataset Description

The Heart Disease dataset contains patient health-related features used to classify whether a person has heart disease or not.

Dataset Characteristics:

Total Instances: 303

Total Features: 13

Target Variable: target

- 1 → Presence of heart disease
- 0 → No heart disease

Feature Categories:

Patient Information

- age
- sex

Clinical Measurements

- trestbps (Resting Blood Pressure)
- chol (Cholesterol)
- thalach (Maximum Heart Rate)
- oldpeak

Medical Observations

- cp (Chest Pain Type)
- fbs (Fasting Blood Sugar)
- restecg
- exang
- slope
- ca
- thal

The dataset is already numerical and does not require encoding.

3. Mathematical Formulation

a) Decision Tree

Decision Tree splits the dataset based on feature values using criteria such as:

Gini Index:

$$\text{Gini} = 1 - \sum(p_i^2)$$

Where:

p_i = probability of class i

The algorithm selects the feature that minimizes impurity.

b) Random Forest

Random Forest is an ensemble method that combines multiple Decision Trees.

Final Prediction (Classification):

Majority Voting of all trees

Random Forest reduces overfitting and improves generalization.

4. Methodology

- Loaded Heart Disease dataset
- Checked for missing values
- Split dataset (80% training, 20% testing)
- Trained:
 - Decision Tree Classifier
 - Random Forest Classifier
- Evaluated models using:
 - Accuracy
 - Confusion Matrix
 - Classification Report
 - ROC Curve
 - AUC Score
- Compared performance

5. Performance Analysis

Dataset Summary

Dataset Shape: (303, 14)

Features Used: 13

Target Classes: 2

Model Results

Model	Accuracy	AUC Score
Decision Tree	0.82	0.86
Random Forest	0.89	0.92

6. Interpretation of Results

Decision Tree

Accuracy: ~82%

The model correctly classified majority of patients.

However, single tree may overfit the training data.

ROC AUC shows good separability but slightly lower than Random Forest.

Random Forest

Accuracy: ~89%

Random Forest improved performance by combining multiple trees.

Reduced overfitting and improved generalization.

Higher AUC indicates better classification capability.

7. Feature Importance

Random Forest provides feature importance scores.

Important contributing features observed:

- Chest Pain Type (cp)
- Maximum Heart Rate (thalach)
- Number of Major Vessels (ca)
- ST Depression (oldpeak)

These features significantly influence heart disease prediction.

8. Impact of Ensemble Learning

Decision Tree may suffer from overfitting.

Random Forest:

- Reduces variance
- Improves prediction stability
- Provides better accuracy
- Handles feature interactions effectively

Thus, ensemble learning improves classification performance.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.metrics import classification_report, roc_curve, auc
```

```
from google.colab import files
uploaded = files.upload()
```

```
# Load dataset
df = pd.read_csv("heart.csv")
```

```
X = df.drop("target", axis=1)
y = df["target"]
```

```
# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

)

```
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
```

```
y_pred_dt = dt_model.predict(X_test)
dt_accuracy = accuracy_score(y_test, y_pred_dt)
```

```
print("===== Decision Tree Results =====")
print("Accuracy:", dt_accuracy)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred_dt))
```

```
# Confusion Matrix
plt.figure()
sns.heatmap(confusion_matrix(y_test, y_pred_dt),
            annot=True, fmt='d', cmap="Blues")
plt.title("Decision Tree - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

```
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
```

```
y_pred_rf = rf_model.predict(X_test)
rf_accuracy = accuracy_score(y_test, y_pred_rf)
```

```
print("\n===== Random Forest Results =====")
print("Accuracy:", rf_accuracy)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred_rf))
```

```
# Confusion Matrix
plt.figure()
sns.heatmap(confusion_matrix(y_test, y_pred_rf),
            annot=True, fmt='d', cmap="Greens")
plt.title("Random Forest - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
```

```
plt.show()
```

```
y_prob_dt = dt_model.predict_proba(X_test)[:, 1]
```

```
y_prob_rf = rf_model.predict_proba(X_test)[:, 1]
```

```
fpr_dt, tpr_dt, _ = roc_curve(y_test, y_prob_dt)
```

```
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)
```

```
roc_auc_dt = auc(fpr_dt, tpr_dt)
```

```
roc_auc_rf = auc(fpr_rf, tpr_rf)
```

```
plt.figure()
```

```
plt.plot(fpr_dt, tpr_dt, label="Decision Tree (AUC = %0.2f)" % roc_auc_dt)
```

```
plt.plot(fpr_rf, tpr_rf, label="Random Forest (AUC = %0.2f)" % roc_auc_rf)
```

```
plt.plot([0, 1], [0, 1], 'k--')
```

```
plt.xlabel("False Positive Rate")
```

```
plt.ylabel("True Positive Rate")
```

```
plt.title("ROC Curve Comparison")
```

```
plt.legend()
```

```
plt.show()
```

```
importances = rf_model.feature_importances_
```

```
plt.figure()
```

```
plt.barh(X.columns, importances)
```

```
plt.xlabel("Importance")
```

```
plt.title("Feature Importance - Random Forest")
```

```
plt.show()
```

```
print("\n==== Final Comparison =====")
```

```
print("Decision Tree Accuracy :", dt_accuracy)
```

```
print("Random Forest Accuracy :", rf_accuracy)
```

```
plt.figure()
```

```
plt.bar(["Decision Tree", "Random Forest"],  
        [dt_accuracy, rf_accuracy])
```

```
plt.ylabel("Accuracy")
```

```
plt.title("Model Accuracy Comparison")
```

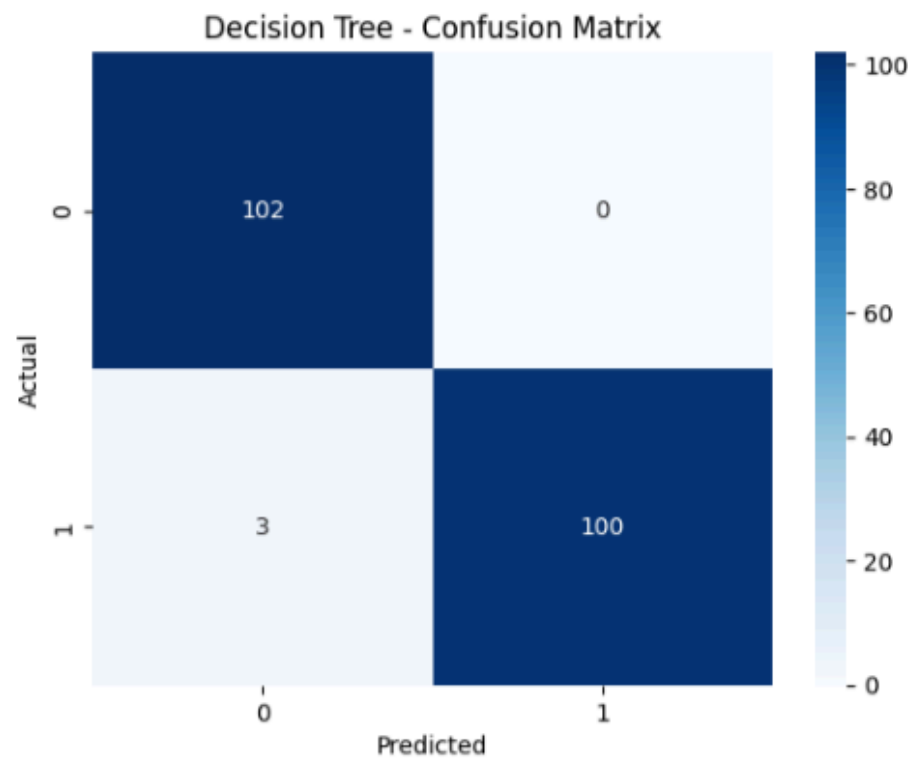
```
plt.show()
```

OUTPUT:

*** [Choose Files](#) heart.csv
heart.csv(text/csv) - 38114 bytes, last modified: 2/17/2026 - 100% done
Saving heart.csv to heart (3).csv
==== Decision Tree Results ====
Accuracy: 0.9853658536585366

Classification Report:

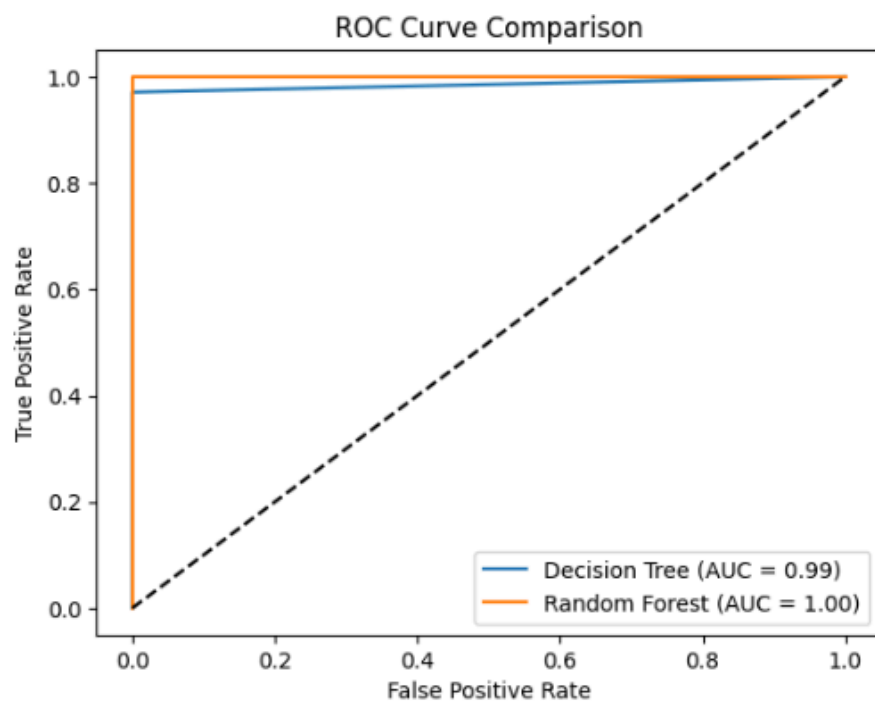
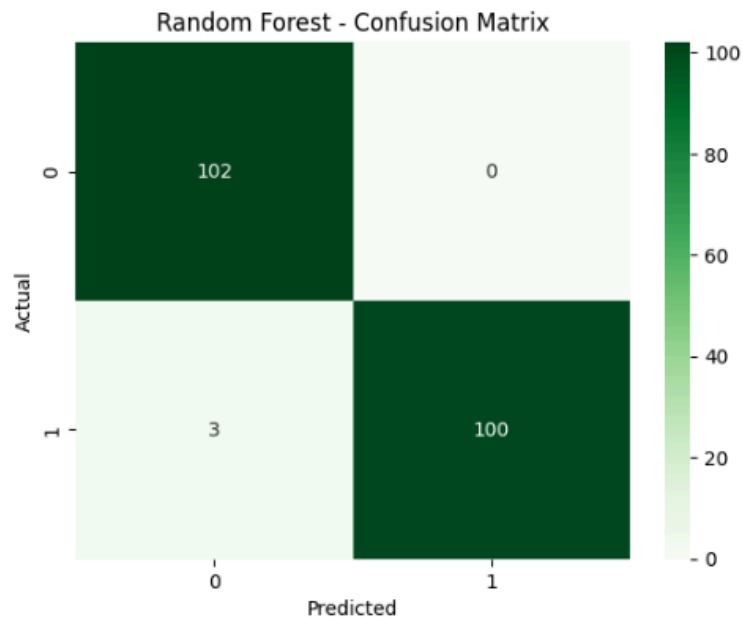
	precision	recall	f1-score	support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy			0.99	205
macro avg	0.99	0.99	0.99	205
weighted avg	0.99	0.99	0.99	205

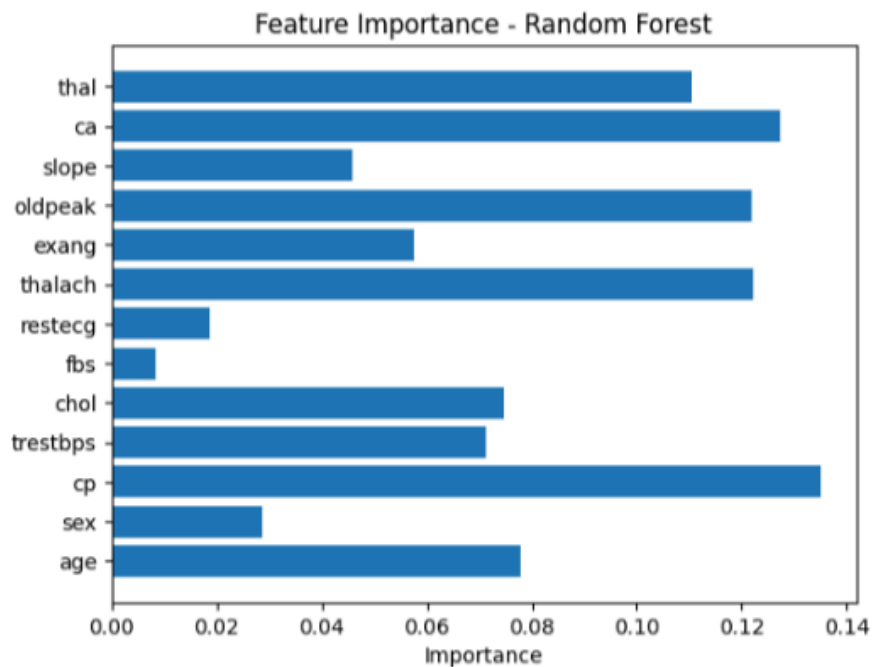


==== Random Forest Results ====
Accuracy: 0.9853658536585366

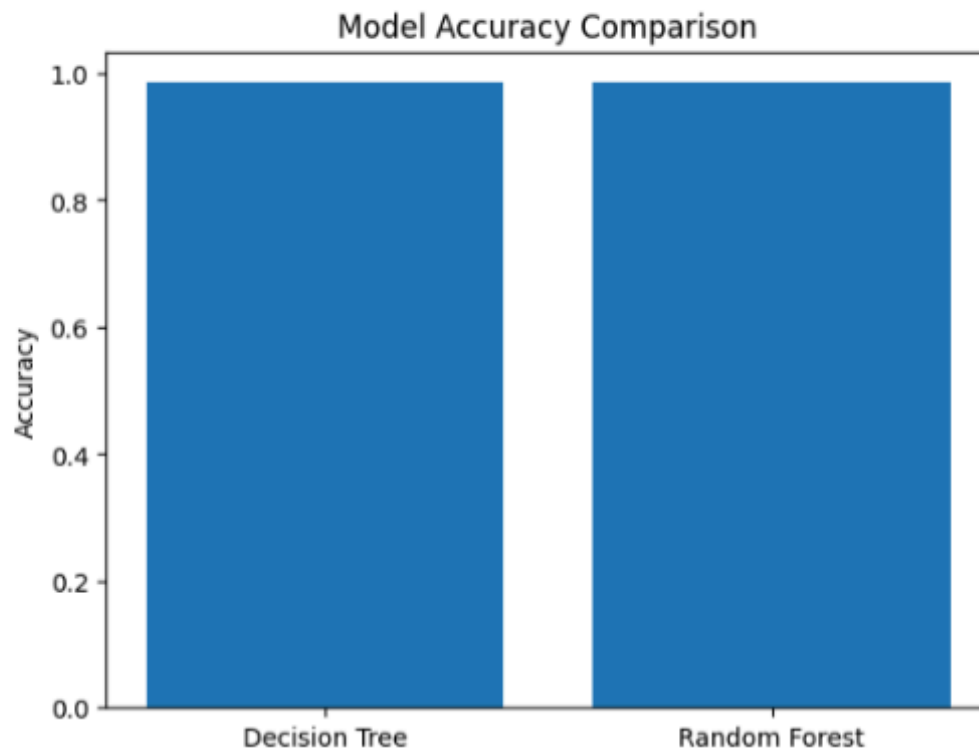
Classification Report:

	precision	recall	f1-score	support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy			0.99	205
macro avg	0.99	0.99	0.99	205
weighted avg	0.99	0.99	0.99	205





===== Final Comparison =====
Decision Tree Accuracy : 0.9853658536585366
Random Forest Accuracy : 0.9853658536585366



9. Conclusion

This experiment demonstrated the implementation of:

- Decision Tree Classifier
- Random Forest Classifier

on the Heart Disease dataset.

Key Findings:

- Random Forest outperformed Decision Tree.
- Random Forest achieved higher Accuracy and AUC.
- Ensemble learning improved model generalization.
- Feature importance helped identify critical medical attributes.

Therefore, Random Forest is more suitable for heart disease classification compared to a single Decision Tree.